San Jose Police Call Cancellation and Response EDA Report

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Objective: Investigate whether high-priority police calls are associated with increased cancellation rates of lower-priority calls due to resource starvation.

Data Overview and Preparation:

Overview

The dataset consists of police calls for service from San Jose Open Data Portal, spanning from 2016 to present. Each record represents a dispatched call and includes 17 key variables with many of major importance such as PRIORITY, CALL_TYPE, FINAL_DISPO, CDTS, and many more.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	CDTS	EID	START_DATE	CALL_NUMBE	PRIORI	REPORT_DATE	OFFENSE_DATE	OFFENSE_TIME	CALLTYPE_CO	CALL_TYPE	FINAL_DISPO	FINAL_DISPO	ADDRESS	CITY	STATE
2	20250102	1E+07	1/1/2025 0:00	P250010001	3	1/1/2025 0:00	1/1/2025 0:00	0:00:02	602PC	TRESPASSING	N	No report required; d	Not a valid geographical location in	San Jose	CA
3	20250101	1E+07	1/1/2025 0:00	P250010006	2	1/1/2025 0:00	1/1/2025 0:00	0:02:34	1057	FIREARMS DISCHARGED	CAN	Canceled	[1700]-[1800] OLIVETREE DR	San Jose	CA
4	20250101	1E+07	1/1/2025 0:00	P250010007	4	1/1/2025 0:00	1/1/2025 0:00	0:03:38	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	MALABAR DR & WESTBORO DR	San Jose	CA
5	20250101	1E+07	1/1/2025 0:00	P250010008	3	1/1/2025 0:00	1/1/2025 0:00	0:04:08	1033A	ALARM, AUDIBLE	CAN	Canceled	[200]-[300] DEVCON DR	San Jose	CA
6	20250101	1E+07	1/1/2025 0:00	P250010009	2	1/1/2025 0:00	1/1/2025 0:00	0:04:36	415A	DISTURBANCE, FIGHT	N	No report required; d	[0]-[100] GEBHART AV	San Jose	CA
7	20250101	1E+07	1/1/2025 0:00	P250010010	4	1/1/2025 0:00	1/1/2025 0:00	0:05:15	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[1500]-[1600] BERRYESSA RD	San Jose	CA
8	20250101	1E+07	1/1/2025 0:00	P250010011	2	1/1/2025 0:00	1/1/2025 0:00	0:06:50	1057	FIREARMS DISCHARGED	CAN	Canceled	HOFFMAN CT & BLOSSOM HILL RD	San Jose	CA
9	20250101	1E+07	1/1/2025 0:00	P250010012	2	1/1/2025 0:00	1/1/2025 0:00	0:07:23	1057	FIREARMS DISCHARGED	CAN	Canceled	MERRILL DR & CAMDEN AV	San Jose	CA
10	20250101	1E+07	1/1/2025 0:00	P250010013	2	1/1/2025 0:00	1/1/2025 0:00	0:07:58	1057	FIREARMS DISCHARGED	CAN	Canceled	[700]-[800] CALERO AV	San Jose	CA
11	20250101	1E+07	1/1/2025 0:00	P250010014	4	1/1/2025 0:00	1/1/2025 0:00	0:08:13	415M	DISTURBANCE, MUSIC	DUPNCAN	No Disposition	[1200]-[1300] FLICKINGER AV	San Jose	CA
12	20250101	1E+07	1/1/2025 0:00	P250010015	2	1/1/2025 0:00	1/1/2025 0:00	0:08:46	1057	FIREARMS DISCHARGED	CAN	Canceled	[1500]-[1600] BORDELAIS DR	San Jose	CA
13	20250101	1E+07	1/1/2025 0:00	P250010016	4	1/1/2025 0:00	1/1/2025 0:00	0:09:13	415FC	DISTURBANCE, FIRECRACKE		No Disposition	HATTON ST & HEMLOCK AV	San Jose	CA
14	20250101	1E+07	1/1/2025 0:00	P250010017	4	1/1/2025 0:00	1/1/2025 0:00	0:11:04	415M	DISTURBANCE, MUSIC	CAN	Canceled	[1200]-[1300] SHERMAN ST	San Jose	CA
15	20250101	1E+07	1/1/2025 0:00	P250010018	3	1/1/2025 0:00	1/1/2025 0:00	0:11:09	SUSCIR	SUSPICIOUS CIRCUMSTANC	CAN	Canceled	[200]-[300] BURNING TREE DR	San Jose	CA
16	20250101	1E+07	1/1/2025 0:00	P250010019	4	1/1/2025 0:00	1/1/2025 0:00	0:11:21	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[100]-[200] HOULTON CT	San Jose	CA
17	20250101	1E+07	1/1/2025 0:00	P250010020	4	1/1/2025 0:00	1/1/2025 0:00	0:11:36	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[900]-[1000] FURLONG DR	San Jose	CA
18	20250101	1E+07	1/1/2025 0:00	P250010021	4	1/1/2025 0:00	1/1/2025 0:00	0:12:40	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[1700]-[1800] OLIVETREE DR	San Jose	CA
19	20250101	1E+07	1/1/2025 0:00	P250010022	4	1/1/2025 0:00	1/1/2025 0:00	0:13:05	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	GRIDLEY ST & JOANNE AV	San Jose	CA
20	20250101	1E+07	1/1/2025 0:00	P250010023	1	1/1/2025 0:00	1/1/2025 0:00	0:13:33	246	SHOOTING INTO OCCP VEH	R	Report taken	[2000]-[2100] CURTNER AV	San Jose	CA
21	20250101	1E+07	1/1/2025 0:00	P250010024	4	1/1/2025 0:00	1/1/2025 0:00	0:14:32	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[5300]-[5400] BIRCH GROVE DR	San Jose	CA
22	20250101	1E+07	1/1/2025 0:00	P250010025	2	1/1/2025 0:00	1/1/2025 0:00	0:15:42	415	DISTURBANCE	N	No report required; d	[900]-[1000] S 6TH ST	San Jose	CA
23	20250101	1E+07	1/1/2025 0:00	P250010026	4	1/1/2025 0:00	1/1/2025 0:00	0:15:47	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[1800]-[1800] STOKES ST	San Jose	CA
24	20250101	1E+07	1/1/2025 0:00	P250010027	3	1/1/2025 0:00	1/1/2025 0:00	0:16:07	602PC	TRESPASSING	N	No report required; d	[4900]-[5000] ALMADEN EX	San Jose	CA
25	20250101	1E+07	1/1/2025 0:00	P250010028	4	1/1/2025 0:00	1/1/2025 0:00	0:17:13	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	COMMODORE DR & N KING RD	San Jose	CA
26	20250101	1E+07	1/1/2025 0:00	P250010029	4	1/1/2025 0:00	1/1/2025 0:00	0:17:43	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	MCCREERY AV & LAVONNE AV	San Jose	CA
27	20250101	1E+07	1/1/2025 0:00	P250010030	2	1/1/2025 0:00	1/1/2025 0:00	0:19:27	WELCK	WELFARE CHECK	R	Report taken	[700]-[800] RIVER PARK DR	San Jose	CA
28	20250101	1E+07	1/1/2025 0:00	P250010032	4	1/1/2025 0:00	1/1/2025 0:00	0:21:29	415FC	DISTURBANCE, FIRECRACKE	CAN	Canceled	[900]-[900] S 10TH ST	San Jose	CA
29	20250101	1E+07	1/1/2025 0:00	P250010034	4	1/1/2025 0:00	1/1/2025 0:00	0:22:18	415M	DISTURBANCE, MUSIC	CAN	Canceled	[2700]-[2800] BERRYESSA RD	San Jose	CA
30	20250101	1E+07	1/1/2025 0:00	P250010035	4	1/1/2025 0:00	1/1/2025 0:00	0:23:00	415FC	DISTURBANCE, FIRECRACKE		No Disposition	[3000]-[3100] HUFF AV	San Jose	CA

Figure 1: Raw Dataset Sample

Data Processing

Several preprocessing steps were performed:

1. Convert timestamp fields CDTS into datetime format
2. Standardizing categorical fields and creating indicator variables (e.g. is_p1, is_p3_p4, etc.)
3. Aggregating into 5-minute bins to analyze system-level behavior

Each time window includes: Total Calls, # of P1 calls, # of P3/P4 calls, # of canceled P3/P4 calls, and cancellation rates. [*Ref. Figure 2*]

	day	total_calls
0	2022-01-01	794
1	2022-01-02	726
2	2022-01-03	804
3	2022-01-04	796
4	2022-01-05	848

Figure 2a: Aggregated Daily Call Volume Table

	hour	total_calls
0	0	22277
1	1	17742
2	2	13632
3	3	10565
4	4	8718

Figure 2b: Aggregated Hourly Call Volume Table

	PRIORITY	total_calls	canceled_calls	canceled_p3_p4	cancel_rate
0	1	12990	650	0	0.050038
1	2	129095	20920	0	0.162051
2	3	147766	40922	40922	0.276938
3	4	44466	17160	17160	0.385913
4	5	28460	232	0	0.008152
5	6	53572	520	0	0.009707

Figure 2c: Aggregated Call Volume and Cancellation Rate by Priority Table

Descriptive Plots

General Call Volume Statistics

In *Figure 3*, we notice that call volume would roughly remain the same and consistent throughout the years. Certain outliers appeared in the data such as January 2022 and 2024 having very high/very low call volumes.

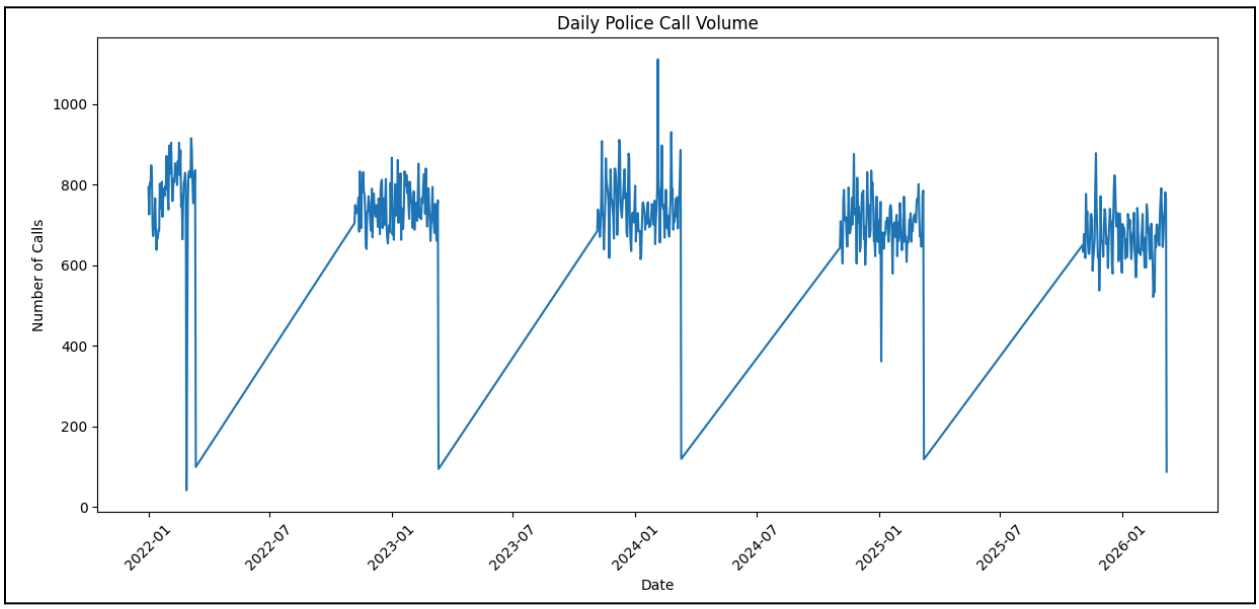

Figure 3: SJPD Daily Call Volume (2022–2026)

Call Volume based on Priority/Hour of day

In *Figure 4*, it can be noted that Priority 2 and 3 measures are the majority of call volume for most days.

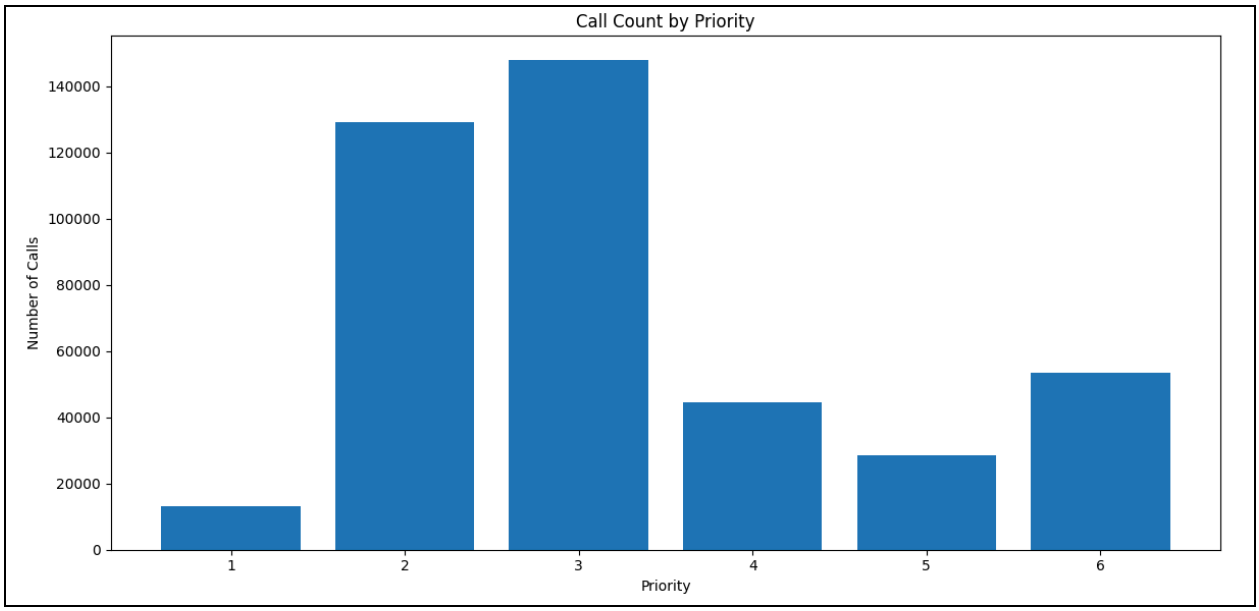

Figure 4: Total Call Volume by Priority Level

In *Figure 5*, for most days there is an uptick in police calls around 4 pm which will typically fall off around 12 am

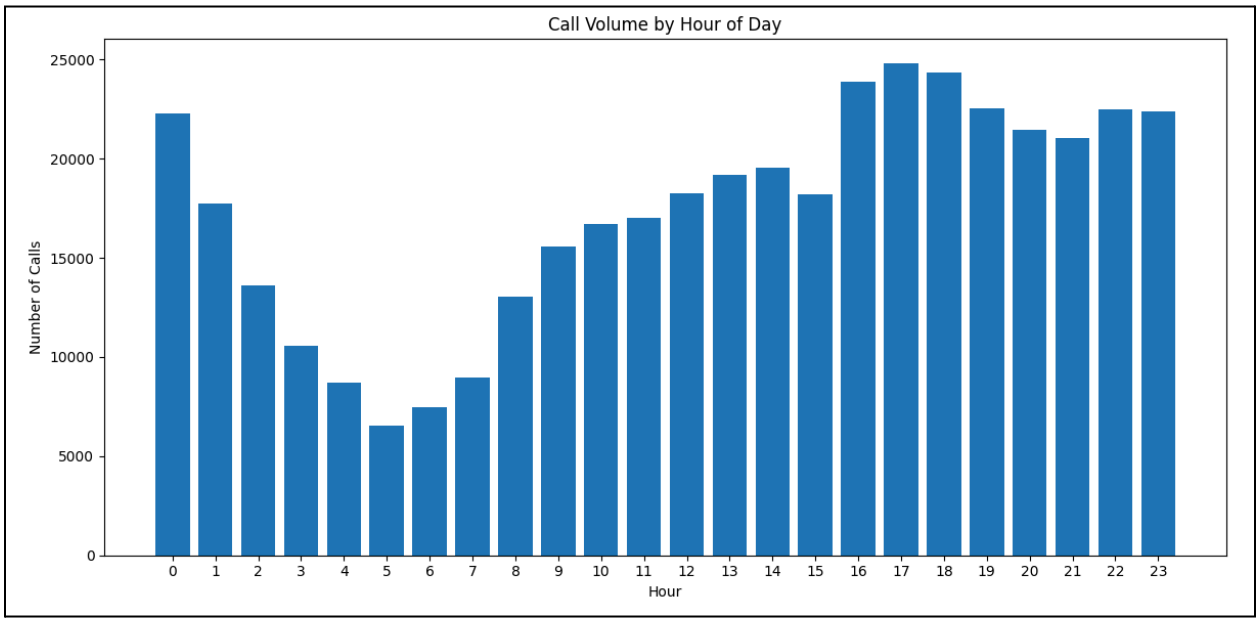

Figure 5: Total Call Volume by Hour of Day

Day of Week and Hour Heatmap

In *Figure 6*, throughout the week, most call volume seems to increase closer to 4 pm and typically trends the same from Monday to Saturday. But from Saturday to Sunday there seems to be a greater uptick in police calls closer to midnight.

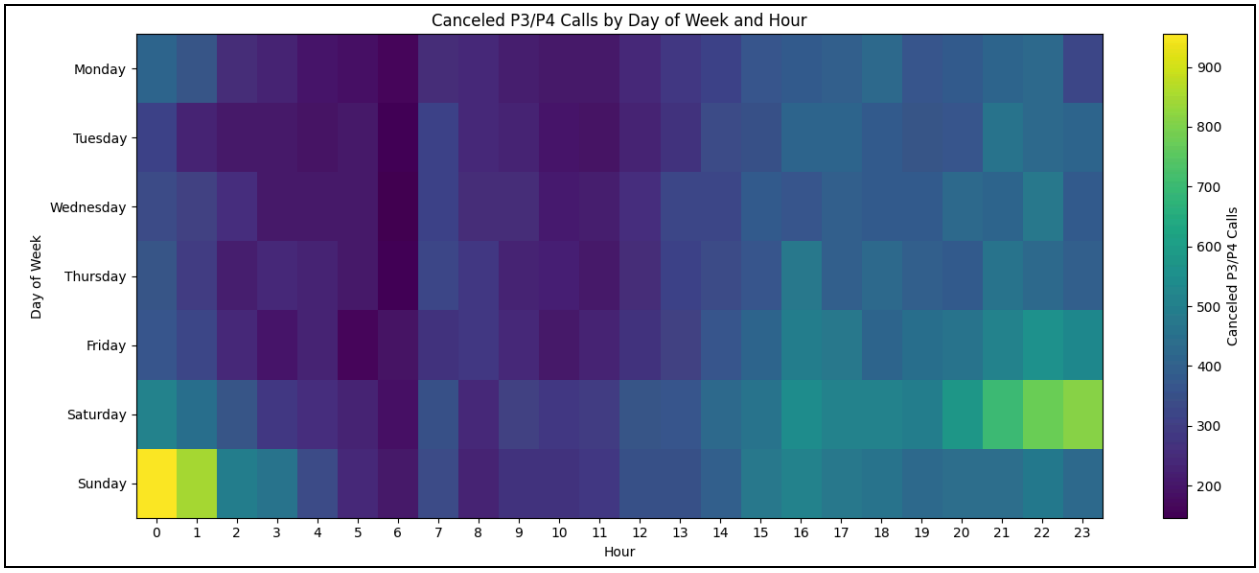

Figure 6: Canceled Priority 3 and 4 Calls by Day and Hour

Monthly/Yearly Summary

Figure 7a and 7b give a look at the cancellation rate from 2022 through early 2026. They show that the average cancellation rate for Priority 3 and Priority 4 calls has been increasing over the past years.

- 2022: The baseline cancellation rate for P3/P4 calls sat at approximately 27.2%.
- 2023: The rate increased to 29.2%.
- 2024: The rate jumps to 31.8%.
- 2025 & 2026: The rate stabilized between 31.4% and 32.0%.

While there are natural month-to-month changes in call volume, the jump to a higher threshold suggests a recent issue or change in policies that could have caused the rates to increase.

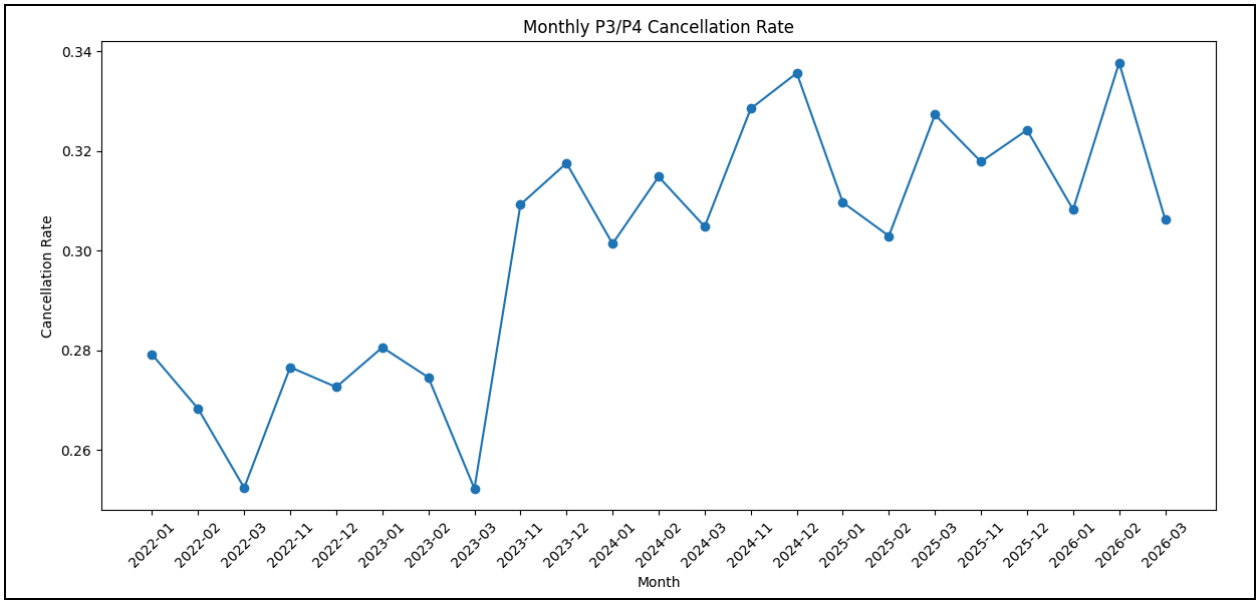

Figure 7a: Monthly P3/P4 Cancellation Rate

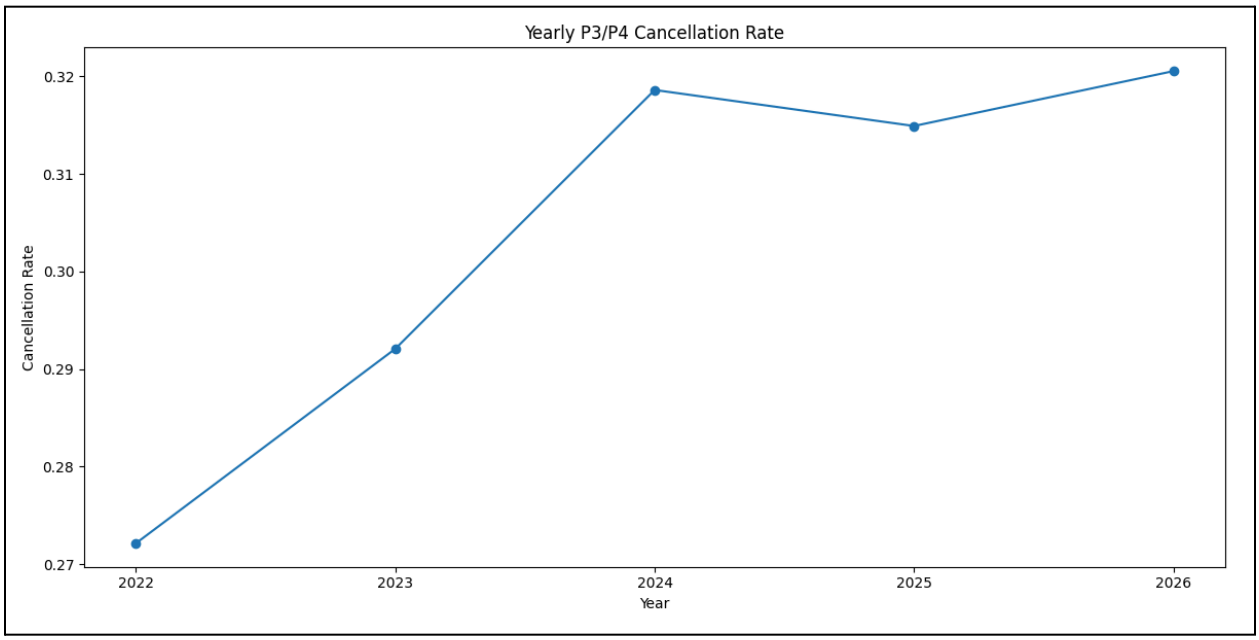

Figure 7b: Yearly P3/P4 Cancellation Rate

Canceled Calls Comparison

Based on *Figure 7 and 8* and the information below:

- Average P3/P4 cancellation rate when there is at least one P1 call: 0.29723155348512376
- Average P3/P4 cancellation rate when there are no P1 calls: 0.2929400682057249
- Correlation between P1 calls and canceled P3/P4 calls: -0.0024170911404780776

We can claim that there is little correlation between Priority 1 calls causing Priority 3 and 4 calls to be cancelled. This is because the average cancellation rate of P3 and P4 calls during and not during a P1 call remains the same in both instances (roughly 40% cancellation rate).

	has_p1	bins	avg_p1_calls	avg_canceled_p3_p4	avg_cancel_rate_p3_p4
0	False	131332	0.000000	0.404844	0.292940
1	True	12366	1.050461	0.397299	0.297232

Figure 8: P3/P4 Cancellation Rate Comparison (With vs. Without P1 Calls)

	p1_bin	num_bins	avg_canceled_p3_p4	avg_cancel_rate_p3_p4	median_cancel_rate_p3_p4
0	0	131332	0.404844	0.292940	0.0
1	1	11773	0.395651	0.296959	0.0
2	2	564	0.437943	0.312412	0.0
3	3	27	0.296296	0.122727	0.0
4	4+	2	0.000000	NaN	NaN

Figure 9: P3/P4 Cancellation Rate Based on Simultaneous P1 Call Volume

Case Study: December 24-25, 2025

We took a look at a specific day in our dataset where there was a massive spike in call volume to perform a breakdown at the types of calls that were potentially causing the system to experience stress.

	time_bin	total_calls	p1_calls	p3_p4_calls	canceled_p3_p4	all_canceled	cancel_rate_p3_p4	has_p1	p1_bin
125583	2025-12-24 04:30:00	17	0	17	17	17	1.000000	False	0
125845	2025-12-25 04:10:00	17	1	16	15	15	0.937500	True	1
72581	2024-02-04 16:25:00	19	1	17	13	13	0.764706	True	1
7831	2022-02-01 00:15:00	13	0	13	11	11	0.846154	False	0
37037	2023-01-22 00:20:00	16	0	12	10	11	0.833333	False	0
125836	2025-12-25 03:25:00	13	0	12	10	10	0.833333	False	0
125809	2025-12-25 00:50:00	12	0	11	10	10	0.909091	False	0
96058	2025-01-01 01:25:00	10	0	10	10	10	1.000000	False	0
22882	2022-11-26 21:15:00	10	0	10	9	9	0.900000	False	0
94305	2024-12-25 01:55:00	10	0	9	9	9	1.000000	False	0
109890	2025-02-25 22:35:00	10	0	9	9	9	1.000000	False	0
19411	2022-11-13 01:35:00	11	1	8	8	8	1.000000	True	1
63923	2024-01-01 02:25:00	13	0	11	8	8	0.727273	False	0
62135	2023-12-24 23:05:00	12	0	8	8	8	1.000000	False	0
54769	2023-11-26 00:10:00	11	0	10	8	8	0.800000	False	0
71371	2024-01-30 21:55:00	11	0	8	8	9	1.000000	False	0
125882	2025-12-25 08:10:00	11	0	8	8	8	1.000000	False	0
141977	2026-03-01 04:00:00	11	0	9	8	8	0.888889	False	0
37034	2023-01-22 00:05:00	10	0	9	8	8	0.888889	False	0
65608	2024-01-07 21:30:00	10	0	8	8	10	1.000000	False	0

Figure 10: Absolute Busiest 5-Minute Call Windows (2022–2026)

We observed that Wednesday, December 24, 2025 (as ranked at the top in the above image), in particular, was a time when we observed many calls to the system. This specific day stands out to us because it indicates a cancellation rate of 100 percent for the 17 simultaneous P3 and P4 calls experienced during that timeframe.

The spike at the five-minute bin starting at 4:30:00 a.m. covers calls between 4:30:00 to 4:34:59 a.m. Looking back at 4:28:00 a.m., around two minutes before this time bin, we observed a Priority 1 "EXPLOSION" logged at 4:28 a.m.

In the minutes following the explosion, the system was flooded with ALARM calls. We can explain this by the shockwave from the explosion triggering alarm calls to the SJPD rather than actual break-ins.

This case is important because it shows that our 29 percent baseline cancellation rate is often due to cases that resolve themselves or, in this case, false alarms, rather than crimes the police department can be dispatched to.

Case study window: 2025-12-24 04:15:00 to 2025-12-24 04:45:00

	CDTS	PRIORITY	CALL_TYPE	FINAL_DISPO	ADDRESS
366922	2025-12-24 04:19:09	3	ALARM	No report required; dispatch record only	[3600]-[3700] PEARL AV
366923	2025-12-24 04:21:02	3	DISTURBANCE	No report required; dispatch record only	[1800]-[1800] STOKES ST
366924	2025-12-24 04:21:07	2	WELFARE CHECK	Report taken	BURTON AV & N 1ST ST
366925	2025-12-24 04:21:38	3	ALARM, AUDIBLE	No report required; dispatch record only	[1000]-[1100] FOXWORTHY AV
366926	2025-12-24 04:23:40	3	TRAFFIC HAZARD	No Disposition	[1200]-[1300] THORNBURY LN
366927	2025-12-24 04:27:06	3	ALARM, AUDIBLE	Canceled	[2100]-[2200] MONTEREY RD
366928	2025-12-24 04:27:20	3	ALARM, AUDIBLE	Canceled	[900]-[1000] OAKLAND RD
366929	2025-12-24 04:28:00	1	EXPLOSION	No report required; dispatch record only	ALMADEN EX & IRONWOOD DR
366930	2025-12-24 04:29:10	3	ALARM, AUDIBLE	Canceled	[0]-[100] KEYES ST
366931	2025-12-24 04:31:03	3	ALARM, AUDIBLE	Canceled	[1600]-[1600] SARATOGA AV
366932	2025-12-24 04:31:08	3	ALARM, SILENT	Canceled	[600]-[700] SARATOGA AV
366933	2025-12-24 04:31:20	3	ALARM, AUDIBLE	Canceled	[800]-[900] S WINCHESTER BL
366934	2025-12-24 04:33:20	3	ALARM, AUDIBLE	Canceled	[1500]-[1600] MOORPARK AV
366935	2025-12-24 04:33:24	3	ALARM, AUDIBLE	Canceled	[3000]-[3100] MONTEREY RD
366936	2025-12-24 04:33:27	3	ALARM, AUDIBLE	Canceled	[1600]-[1700] LUCRETIA AV
366937	2025-12-24 04:33:30	3	ALARM, SILENT	Canceled	[3200]-[3200] WATER ST

Figure 11: 30-Minute Demand Spike Breakdown (Dec 24, 2025)

Conclusions and Community Impact

Our data conclusively prove that the San Jose Police dispatch system does not experience "breaking points" in its system in which it is forced to cancel lower-priority calls to deal with higher-priority ones instead.

Instead, we are focusing on the growing baseline cancellation rate for low-priority calls across the board.

While our case study shows a good example of there always being false calls sent to the system, we can pivot and look for ways to reduce cancellation rates in situations that are not false alarms.

For example, time-sensitive but lower-priority calls like noise complaints and suspicious vehicles are resolved before an officer is dispatched to the scene. This could create frustration among the callers as they did not get the help they needed when they most needed it.

The Real Problem of 29% Baseline

We have two possible ideas as to why the flat 29 percent low-priority call-cancellation rate is apparent.

Without the historical staffing data, a possible improvement could be a transition to more dynamic shift scheduling to align with peaks at certain times of the day, such as after 4:00 p.m. We know that if the police force relies on static scheduling, then misalignment with the actual crime surges would inevitably result in low-priority calls being canceled due to low-staffing issues.

The other idea is that even if the police department had more officers and resources, it is still possible to experience a 29 percent cancellation rate because of the nature of the calls. If there is always a certain percentage of calls that are false alarms or end up resolving themselves in a short period of time, then

having more officers might not entirely resolve the issue. In this case, we can look toward the encouragement of alternate reporting methods or other ways to filter out self-resolving issues from the primary emergency line.

What This Means for SJ Residents

We took a look at how the SJPD could potentially change up their policies in an effort to reduce the baseline cancellation rate. However, the impacts go further than just reducing a number, as there are direct positive impacts it has on the community as well:

- **Peace of Mind**: Our research proves that the dispatch system does not break during major emergencies. This gives citizens peace of mind that help is always available.
- **Data-driven staffing**: We observed a surge of calls after 4:00 p.m. Doing surge shifts instead of static ones possibly helps citizens get an officer before the situation resolves itself.
- **Smarter use of tax dollars**: If we are able to schedule shifts based on predicted stress points, the reliance on expensive overtime pay is reduced and instead allocated more efficiently.
- **Proactive over reactive response**: We can also look toward having a visible police presence ready in areas when and where the call volumes are predicted to ramp up.

By taking steps to allocate resources in a way that matches the actual real world demand of their city, we hope the result would be a community that feels supported by their police officers and a baseline cancellation rate that is driven below the current 29% constant.